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Rigless Water Shut-Off in High CO₂, High-Temperature Mature Deep Wells Using Thru-Tubing Plug and Specialized Cement: A Case Study of Optimized Intervention and Enhanced Recovery

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Abstract

Water production management in mature fields poses a significant challenge, especially in wells impacted by high CO₂, temperatures, and casing deformation. This paper presents a successful rigless Water-Shut-Off (WSO) in a complex mature field well, characterized by high CO₂, temperature exceeding 130°C, and a high inflation ratio requirement for a thru-tubing inflatable plug. The operation was further complicated by casing ovality and a marginal gap between perforations, demanding highly customized and technically sound solution for WSO.

A detailed pre-job analysis was performed, incorporating identification of water entry zone, casing ovality through a high-resolution caliper log to assess casing strength degradation caused by a long-term CO₂ exposure and perforations. The intervention was executed entirely rigless using wireline-deployed tools, starting with the precise placement of an inflatable thru-tubing plug. The plug was inflated and anchored securely, ensuring effective mechanical isolation. ACO₂ resistant, high temperature cement slurry was specially designed for the operation, incorporating temperature-stable retarders to ensure its effective placement and long-term durability. The cement was deployed using a wireline dump bailer and carefully placed above the plug to form a robust zonal barrier.

The application of the integrated solution successfully isolated the water-producing zone, resulting in significantly increased in gas production rates and well productivity by improving wellbore hydraulics with ~98% reduction in water production rate. The elimination of high-water content in a high CO₂ environment, which creates a highly corrosive condition, significantly enhances the life and integrity of the surface facility. The shut-off of high water and CO₂ producing intervals also reduced the risk of scale formation in the wellbore and mitigated the costly scale removal intervention jobs.

This case study highlights the importance of tailored approach for rigless WSO despite multiple operational complexities including high CO₂, elevated temperature, high inflation ratio of thru-tubing plug and casing ovality. Successfully rigless WSO eliminated a huge rig-based workover cost. This rigless water shut-off solution not only restored the hydraulic efficiency of the well but also reduced operational costs and surface facility loads, demonstrating a replicable approach for similar wells in high-CO₂, high-temperature (HT) mature fields.

Introduction

Unwanted water production is one of the most pressing challenges in mature gas fields, particularly in high-permeability sandstone reservoirs where water coning, channeling, or its matrix bypassing can lead to significant gas productivity loss and higher operating costs. This issue is further intensified in reservoirs with high CO₂ concentrations (M Khan 2020) and elevated bottomhole temperatures, which accelerate corrosion and complicate fluid separation at the surface. With the water-to-gas ratio (WGR) has increased over the field life, necessitating effective, economically viable, and less damaging WSO techniques.

Traditional WSO methods rely on full workover operations involving rig-based interventions that, while being effective, are both time consuming and capital intensive. Additionally, such interventions can cause irreversible formation damage due to excessive kill fluid losses, especially in sandstone formations. The development and deployment of rigless WSO techniques, particularly those employing inflatable thru-tubing plugs combined with cement dumping, offer promising alternatives that preserve wellbore hydraulic integrity, minimize deferment, and enhance economic viability.

This paper presents the successful design and implementation of rigless WSO operations in high CO₂ gas wells located in sandstone reservoirs with temperatures exceeding 130°C. It discusses technical screening, tool selection, candidate well analysis, and operational lessons learned. By comparing rig-based and rigless approaches, this study also underscores how rigless WSO methods mitigate reservoir damage and optimize post-WSO wellbore hydraulics. A comprehensive view is provided on how water production control influences plant handling capacities, corrosion risk, and overall asset integrity.

Literature Review

A broad spectrum of WSO techniques has been explored globally, especially in onshore mature gas fields. These include mechanical isolation (bridge plugs, straddles), chemical treatments (gels, resins), cement squeezes, and water shut-off polymers. Bridge plug-based methods have been widely used in vertical wells, especially where the water zone is distinctly separate (Azab 2021). However, these methods often require rig-based operations and are not feasible in wells with tubing restrictions.

Chemical-based WSO methods, particularly polymer gels (e.g., cross-linked polyacrylamide or silicate gels), have shown promise in both matrix and fracture-dominated systems (Taha 2019). Nonetheless, their deployment in HT (>100°C) and high-CO₂ environments is limited by thermal and chemical stability. Cement plug techniques, although traditional, remain effective for zonal isolation when applied with precision, especially in vertical gas wells. Recent advancements in cement slurry design, including latex-modified and CO₂-resistant cements, have further extended their use in harsh environments.

Inflatable thru-tubing isolation plugs have gained significant attention due to their rigless deployment and retrievable/non-retrievable configurations. These plugs, often set via a slickline or an e-line and inflated with a fluid or pressure and serve as robust barriers in wellbores for zonal isolation or plug-and-abandonment (Al-Shahrani 2007). Their adaptability to various casing sizes and ability to pass through narrow restrictions make them ideal for the selective WSO in gas wells.

Selection of Candidate Well

A mature field of Pakistan with more than 150 number of wells having multiple stacked reservoirs was selected for the application of the WSO job. Amongst stacked reservoirs, the bottom most sandstone reservoir is producing high CO₂ gas along with the high water production rate. Handling of the produced water and the high CO₂ also required high-grade surface facilities material which tends to increase the project economics. More than 20 wells were drilled in this reservoir and a significant gas volume is being cumulatively produced from these wells. The production logging job was being conducted to monitor the high-water producing intervals. The parameters like gap between perforation intervals, water/CO₂

production, casing integrity, cement behind casing and water channeling behind the casing were critically examined. By carefully evaluating these factors, Well-X & Well-Y were selected as the most appropriate wells for the WSO technique to optimize resource allocation and ultimately enhance the economic viability and longevity of this valuable gas asset. Table-1 shows the summary of candidate selection results performed for the successful execution of the WSO job.

Table 1—Selection of Candidate Wells for WSO

Well	Well-X	Well-Y	Well-Z
Water Entry Zone Identification	2674.5 - 2677 m (87% Gas) 2686.0 - 2688 m (8% Gas) 2706.5-2714 m (5%Gas/100%Water)	2673 - 2679 m (68% Gas) 2690 - 2693 m (12% Gas / 3% Water) 2702 -2708 m (20% Gas/97% Water)	2681.2-2684.5 m (93% Gas/3% Water) 2691.5-2700 m (7% Gas/97% water)
Cross Flow	No	No	No
Casing Ovality/Integrity	No	No	No
Scale	No	No	Yes (casing ID reduced to 4.88" from 6.094")
Perforation Analysis	Ok	Ok	OK
Compatibility of plugs/Cement	Ok	Ok	OK
Estimated Increment in Gas Rate after WSO (MMSCFD)	~ 29	~ 15	~

For sandstone formations, matrix sensitivity to filtrate and solids makes formation damage due to well killing with the rig-based WSO. Therefore, rigless WSO using inflatable plugs offers a non-invasive method to isolate the watered-out zone with reduced formation exposure. High-CO₂ environment also demands corrosion-resistant materials, especially for downhole plugs and cement additives. Moreover, temperature compatibility of plug elastomers and for the plug setting is crucial to avoid post-setting failures. For the success of any WSO campaign in gas wells, proper consideration of the following key factors is paramount.:

- **Water Entry Zone Identification:** Accurately identifying the source of water ingress (e.g., specific perforations, casing leaks, or water coning) is important. This can be achieved through production logs (PLT), and temperature logs,
- **Wellbore Integrity:** Assessing the condition of the casing, including any deformation, corrosion, or ovality, as these are crucial for selecting appropriate mechanical isolation tools ensuring successful placement of cement. High-resolution caliper logs are essential for this assessment.
- **Reservoir Characteristics:** Permeability, porosity, fluid viscosity, and reservoir pressure influence the choice and design of chemical WSO treatments.
- **Fluid Composition:** The presence of corrosive gases like CO₂ and H₂S, as well as high salinity water, impacts the material selection for plugs and the chemical stability of cement slurries and other injected chemicals.
- **Temperature:** Elevated temperatures significantly affect the performance and stability of both mechanical tools (e.g., seal integrity of plugs) and chemical systems (e.g., setting time of cement, degradation of polymers).
- **Operational Constraints:** Wellbore deviation, depth, and the availability of specialized equipment (e.g., wireline tools, dump bailers) influence the feasibility of different techniques.
- **Economic Viability:** The cost of intervention, including rig time, equipment, and materials, must be weighed against the potential increase in hydrocarbon production and reduction in operational expenses. Rigless operations offer significant cost advantages.

Screening Criteria for WSO Techniques

A screening criterion was developed based on the candidate wells named Well-X & Well-Y, based on the candidate selection criteria mentioned in Table-1. In this criterion, multiple water shut-off techniques like setting of straddle packer, cement squeeze technique, injection of polymer gel and setting of thru-tubing inflatable plugs were critically analyzed. All the WSO techniques were categorized based on factors like formation damage, temperature capacity, CO₂ compatibility, deployment method and cost efficiency. These parameters are critical enough to justify the well economics and successful deployment of a suitable WSO technique.

For this purpose, all the selected WSO techniques were tabulated in Table-2 and analyzed based on the critical factors. It can be clearly seen from Table-2 that among all the mentioned WSO techniques, setting of inflatable plugs to isolate the water producing interval was found to be most useful and cost effective. Also, the thru-tubing inflatable plugs have no impact on formation damage and very much compatible with the high-CO₂ and temperature environment. Moreover, the deployment methods of inflatable plug on a wireline/CT or a slickline is also very convenient. Table-2 summarized the critical evaluation of selection of the most suitable water shut-off technique.

Table 2—Comparison of various WSO techniques

Technique	Formation Damage Risk	Temperature Tolerance	CO ₂ Compatibility	Deployment Method	Cost Efficiency
Straddle Packer	Moderate	High	High	Rig/CT	Moderate
Cement Squeeze	High	Moderate	Low	Rig/CT	Moderate
Polymer Gel Treatment	Moderate	Low to Moderate	Low	CT	Moderate
Inflatable Plug	Low	High	High	CT/E-Line/Slickline	High

Based on the above-mentioned comparison in Table-2, inflatable plug technique was selected for the zonal isolation to improve hydrocarbon productivity, shut-off water producing zone considering high CO₂ and HT well.

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Inflatable Thru-Tubing Plugs: Design and Function

Inflatable plugs consist of a metal mandrel overlaid by a reinforced elastomeric bladder that expands upon the fluid or pressure application. The plug is conveyed through an existing tubing (typically via slickline or an e-line) and inflated using a downhole mechanical plug setting tool once it reaches the target depth. Two variants exist: retrievable (with a bypass in case if there is a cross-flow) and permanent (non-retrievable) types.

In our field application, high-expansion inflatable bridge plugs were selected due to their ability to pass through 2.75" ID tubing and expand up to 8.535" casing ID. These plugs were cement-compatible and temperature-rated up to 150°C. The plug was followed by a dump bailer-based cement placement, which provided a permanent seal without tubing removal.

Despite their advantages, inflatable plugs have several technical and operational hurdles:

- **Inflation Reliability:** Inaccurate inflation can cause the plug under-expansion or premature failure. In high CO₂ wells, elastomer swelling or blistering may occur.
- **Setting Depth Accuracy:** Incorrect setting due to marginal gap between perforations sets can lead to ineffective isolation or plug slippage.

- **Elastomer Selection:** High-CO₂ and HT exposure necessitate premium-grade elastomers to resist degradation.
- **Cement Placement Accuracy:** Dump bailer techniques need calibrated volumes to avoid cement contamination or poor bonding.
- **Cross Flow Risk:** Plug may not properly set or inflated; crossflow can occur around the plug leading to ineffective shut-off.
- **Deployment in Deviated Wells:** High deviation may complicate tool descent and depth control



Figure 1—Figure showing MPBT plug tool string run on wireline

Methodology of Rigless Water Shut-Off in High CO₂ & HT Wells

The methodology employed in this case study for the rigless WSO was meticulously planned and executed to address the severe challenges posed by high CO₂, elevated temperatures, casing ovality, and the need for a high inflation ratio thru-tubing plug. For this purpose, following detailed pre-job analysis was carried out.

Detailed Pre-Job Analysis

Water Entry Zone Identification. A comprehensive analysis was performed to precisely pinpoint the water-entry zone and to identify any crossflow at the plug setting depth. This involved reviewing historical production data, cement bond logs and, crucially, running saturation log, water flow log and production logging tools to determine the water saturation across the perforation intervals, source of water entry and any flow behind the pipe.

Casing Ovality Assessment. The high-resolution caliper log was instrumental in assessing the degree of casing ovality and evaluating the casing strength degradation resulting from long-term exposure to CO₂. This information was critical for designing a thru-tubing plug capable of achieving effective sealing in a deformed casing. Figure-2 showing the results of caliper logs acquired on wireline to assess the degree

of casing deformation/ovality. It can be seen that well X & Y showing the minimal clearance required to set the thru-tubing inflatable plug and successful sealing the bottom zone. However, Well-Z showing the deposition of scale on the walls of tubing and having insufficient clearance required for the setting of the thru-tubing plug.

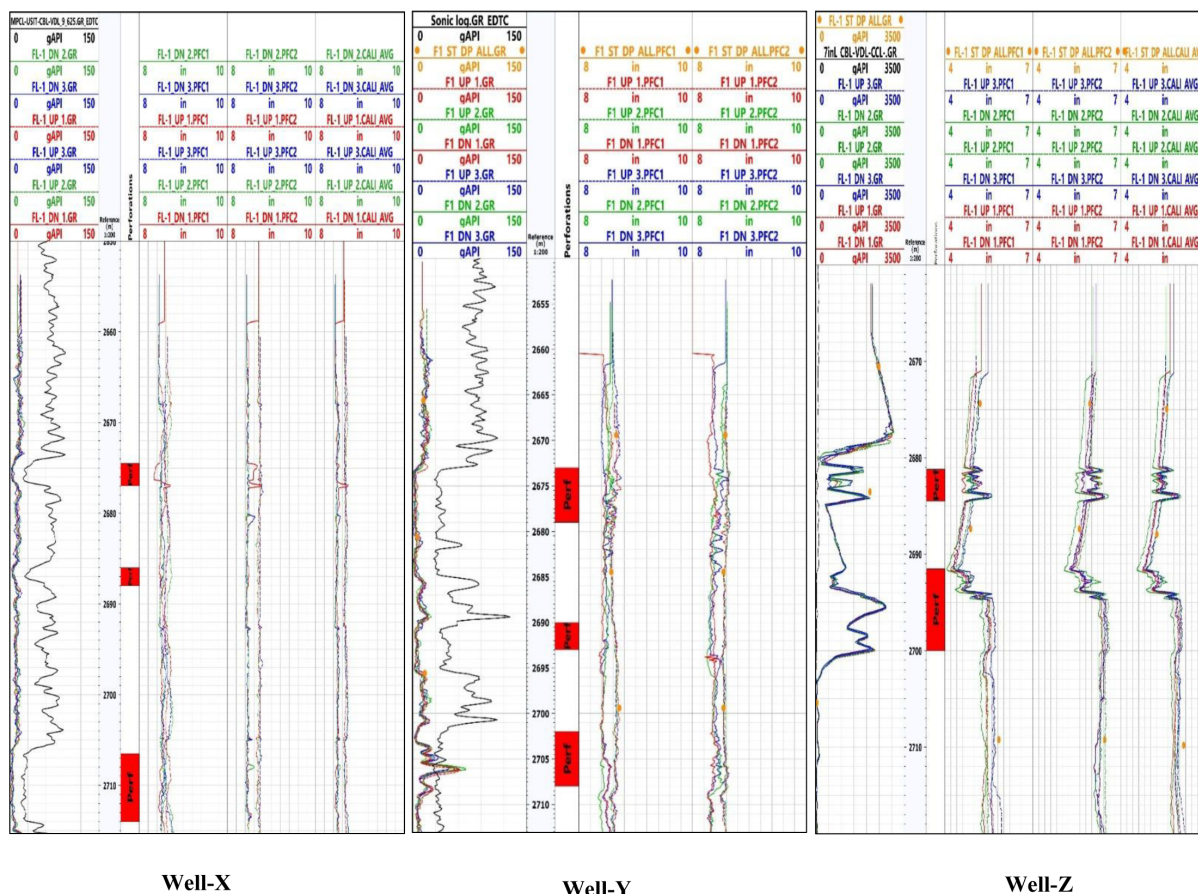


Figure 2—Results of Caliper Log acquired from wireline logging tools

Scale Assessment. Through the high-resolution caliper and gamma ray logs, assessment was performed to determine presence of any scale due to the high-water production across the perforation intervals

Perforation Analysis. The caliper log also helped in analyzing the perforations, especially the marginal gap between them, which would influence the precise placement of the plug and cement.

Wellbore Modelling. Wellbore hydraulics models were developed for each well and matched/validated with the current production scenario. Figure-3 shows the wellbore models developed for Well-X & Y as the WSO is not possible for Well-Z as per the selection criteria mentioned in Table-2. It is clearly observed from Figure-3 that currently Well-X is producing almost 8 MMSCFD with 1500 BPD water production and Well-Y is producing around 15 MMSCFD gas with 2050 BPD water production.

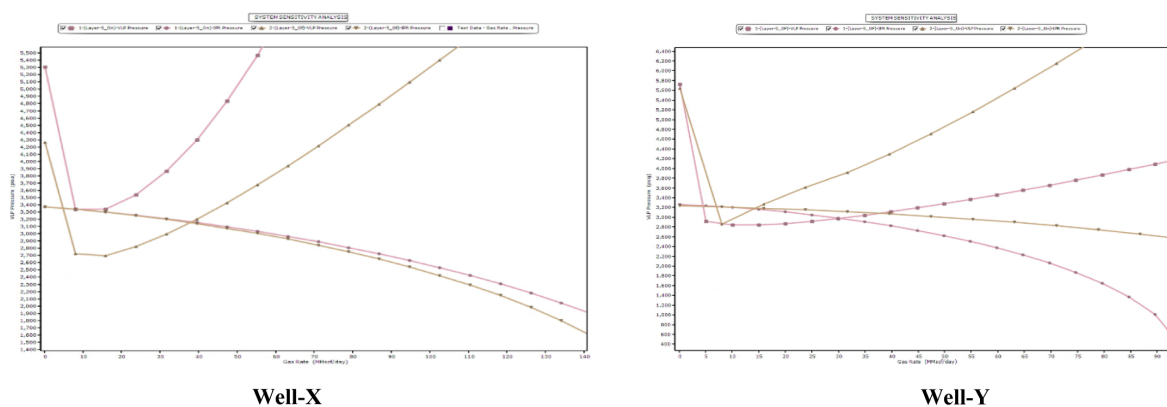


Figure 3—A comparison of estimated gas production rates (pre & post WSO)

After assuming a successful execution of setting of thru-tubing inflatable plugs, estimated gas production rate of Well-X increased up to 37 MMSCFD with reduction in WGR from 187 to zero and 30 MMSCFD from Well-Y with reduction in WGR from 140 to zero. A total of 29 MMSCFD incremental gas rate is observed from Well-X and 15 MMSCFD incremental gas rate from Well-Y is observed from the wellbore models.

Production Logging. Production logging job was performed on each well to identify the zonal contribution of each perforation interval. Figure-4 shows the results of production logging job on Well X, Y & Z. It is clearly observed from figure that the bottom-most interval are high water producing interval and the shut-off is required to improve the wellbore hydraulics and gain incremental gas production.

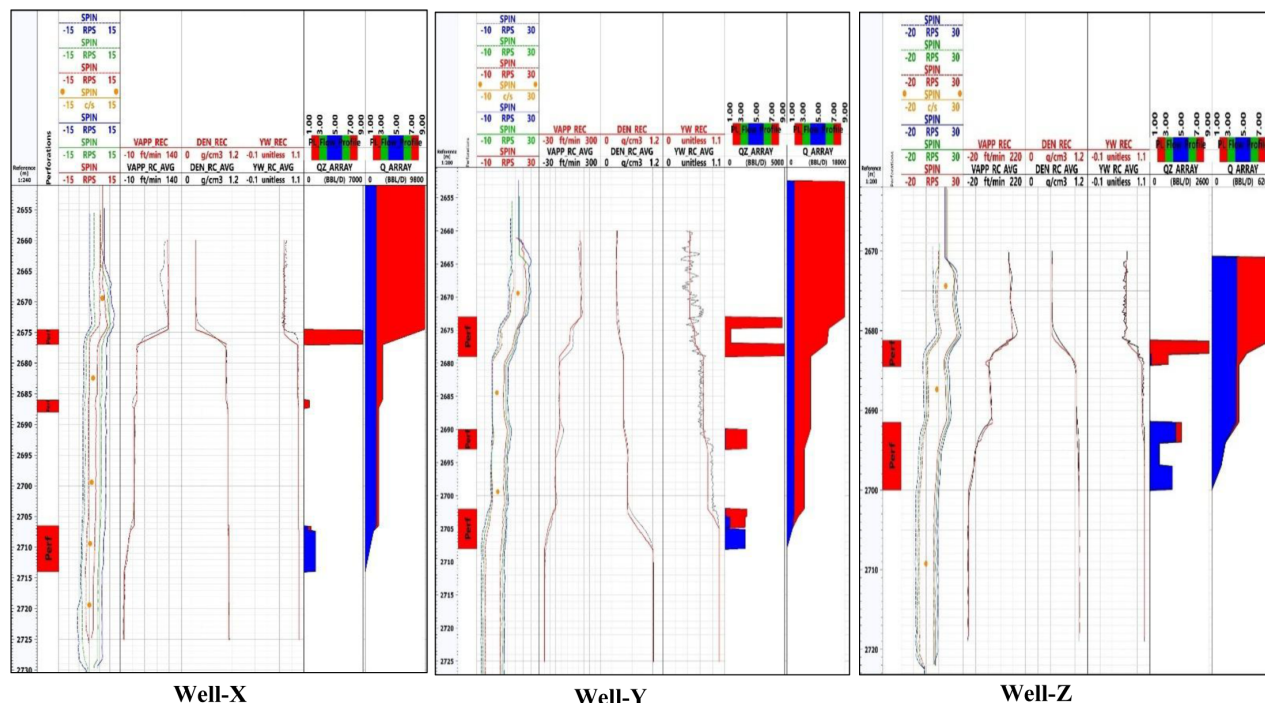


Figure 4—Results of Production Logging

Comparison of Rigless WSO with Traditional Rig-Based Workovers. Sandstone reservoirs, especially with high-permeability streaks, are vulnerable to formation damage during well killing and often leads to irreversible permeability reduction. Rigless WSO techniques using inflatable plugs avoid this by eliminating

the need for well killing, hence preserving the gas production potential. A detailed comparison of rigless and rig-based WSO performed and shown in Table 3.

Table 3—A comparison of Rigless and Rig Based WSO

Feature	Rig-Based WSO	Rigless WSO
Formation Damage Risk	High due to kill mud invasion	Minimal
Operational Simplicity	Complex, multiple runs	Simple – single-run deployment
Tubing Removal	Required	Not Required
Production Loss	High (Due to Well Shut-in during WO)	Minimal
Duration	10 – Days Longer duration due to rig mobilization, setup, and more complex operations	1 – Days Shorter intervention time, faster well restoration
Cost	High (USD 1.5 million)	Low (USD 150K)
Risk	Higher operational risks associated with heavier equipment	Lower operational risks due to simpler procedures
Environmental Impact	Higher environmental impact due to larger equipment and increased logistical needs	Reduced carbon footprint due to shorter operation

Rigless Job Execution

Dummy Run with 2.70" OD Gauge Cutter. After the candidate well selection and detailed pre-job analysis, a systematic wireline conveyed approach was adopted to ensure successful setting of a thru-tubing inflatable plug in a gas well. Dummy run with 2.70" OD gauge cutter and 12 m string length was performed to verify the clearance inside the production tubing and ensure smooth passage to the intended setting depth. This was critical to confirm the absence of any restrictions and avoid tool hang-up/sticking during the actual plug run.

Setting of Thru-Tubing Inflatable Plug (MPBT Plug). A surface function test of the plug setting tool was conducted to verify actuation reliability and inflation response. This involved pressurizing the tool at surface up to 4500 psi while monitoring the setting signature to simulate downhole deployment. Depth accuracy plays a pivotal role in plug setting. Since the physical measurement from surface cannot directly indicate the mid-point of the inflatable element, after connecting the MPBT (Mechanical Plug Back Tool) plug with wireline tool string, length measurements from the Casing Collar Locator (CCL) to the mid of the plug element were derived from manufacturer drawings. This ensures precise anchoring at the desired depth, especially important when targeting a specific water-producing zone for isolation with a narrow depth margin.

The plug was run in-hole and set at the predetermined depth after correlating depth with Cased hole log. The plug setting was confirmed by monitoring wireline tension drop (~60 lbs) and pressure signature during inflation, indicating anchor engagement. Following the plug setting, a blind box was run to physically tag the top of the plug to verify that it has been anchored at the intended depth and had not slipped during inflation.

Standalone setting of the MPBT plug is not sufficient for isolating the desired high-water producing interval in HT and High CO₂ environment, still a secondary barrier is required to support the isolation of the plug. For this purpose, dumping of cement on the MPBT plug was performed. However, due to presence of high CO₂ in the production stream, a specially-designed cement slurry was required to avoid any cement integrity and setting issues.

Cement Recipe Design, Lab Test and Strength. In designing a cement slurry for effective water shut-off in high CO₂ and elevated temperature environments, several key factors were meticulously considered to ensure long-term performance and durability. The cement recipe was tailored to withstand the corrosive effects of CO₂ and the thermal stress encountered in deep reservoirs. Critical considerations included

selecting appropriate cement type, that offers resistance to the thermal degradation and the CO₂ induced corrosion. The slurry's composition was optimized with additives, including retarders to control thickening time at high temperatures, and CO₂ resistance to minimize the risk of carbonation and subsequent cement degradation. Additionally, the water-to-cement ratio was carefully adjusted to maintain the necessary fluidity and compressive strength, while ensuring proper sealing and isolation. The resulting slurry design ensures reliable and prolonged WSO, safeguarding well integrity and minimizing reservoir contamination over time in harsh downhole conditions.

Portland cement systems are used conventionally for isolation in wells. However, portland cement is thermodynamically unstable in CO₂-rich environments and can degrade rapidly upon exposure to CO₂ in the presence of water. Eventually, the compressive strength of the set cement decreases and the permeability and porosity increase, leading to loss of zonal isolation. CO₂ Resistant Cement Slurry combats the risk of cement sheath degradation in the presence of CO₂ environments, while also providing a competent pressure seal to protect against hydrocarbon leaks. The flexibility of this system makes it durable to withstand common cement sheath stresses under different load conditions, perforating, temperature and pressure variations. This system is combined with expanding additives to give flexible and expanding properties, which act to re-establish integrity and recover zonal isolation in the event of damage to the cement matrix and potential CO₂ migration. The results of specialized CO₂ Resistant cement slurry's composition (Pozzolan CO₂ Resistant, Flex Agent + Expandible Agent), rheology and thickening time of cement are tabulated in below tables.

Table 4—Composition of Cement Slurry

Cement Slurry Composition		
Slurry Density	16.20	Lb/gal
Yield	1.75	ft ³ /sk of cement
Mix Fluid	8.314	Gal/sk of cement
Mix Water	4.584	Gal/sk of cement
Solid Vol. Fraction	48.57	%
Slurry Type	CO ₂ Resistant	

Table 5—Rheology of Cement

Surface				Downhole			
Temperature	80 °F			Temperature	190 °F		
rpm	Up (deg)	Down (deg)	Average (deg)	rpm	Up (deg)	Down (deg)	Average (deg)
300	261	261	261	300	229	229	229
200	198	196	197	200	167	166	167
100	125	122	124	100	106	105	106
60	85	83	84	60	75	74	75
30	48	47	48	30	42	41	42
6	27	26	27	6	17	16	17
3	23	22	23	3	14	13	14
10 sec Gel	24 deg - 26 lbf/100ft ²			10 sec Gel	14 deg - 15 lbf/100ft ²		
10 min Gel	48 deg - 51 lbf/100ft ²			10 min Gel	32 deg - 34 lbf/100ft ²		
Rheo. Computed	PV: 243 cP, Ty: 28 lbf/100ft ²			Rheo. Computed	PV: 215 cP, Ty: 21 lbf/100ft ²		

Table 6—Thickening Time of Cement Slurry

Thickening Time of Cement Slurry		
Consistency	Time	Temperature
Motor on	Motor off till	Duration
00:00 hr:mn	01:30 hr:mn	270 °F
30 BC	04:30 hr:mn	270 °F
70 BC	04:32 hr:mn	270 °F
100 BC	04:37 hr:mn	270 °F

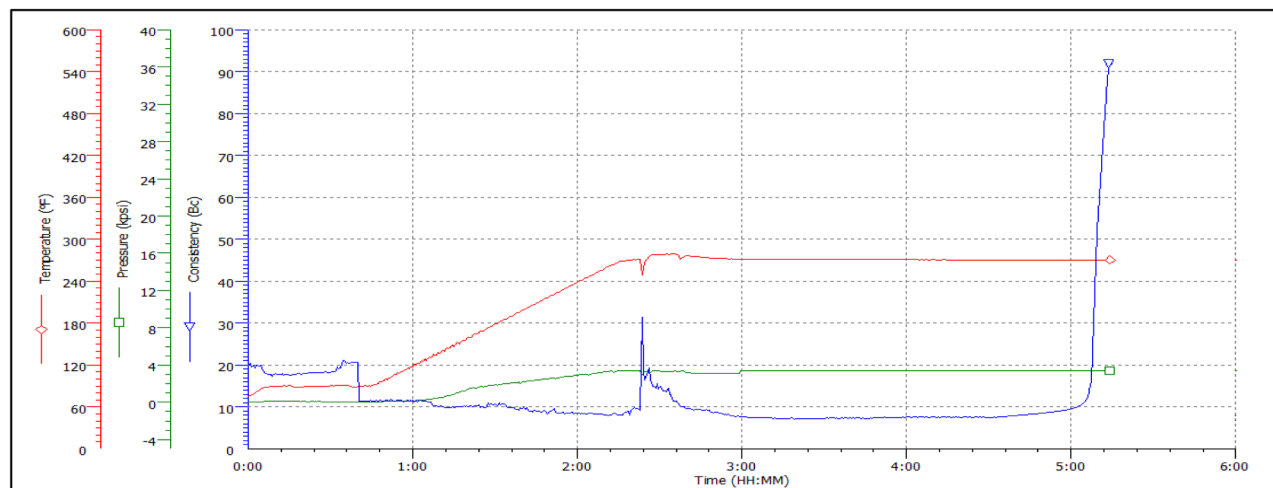


Figure 5—Thickening Time Chart

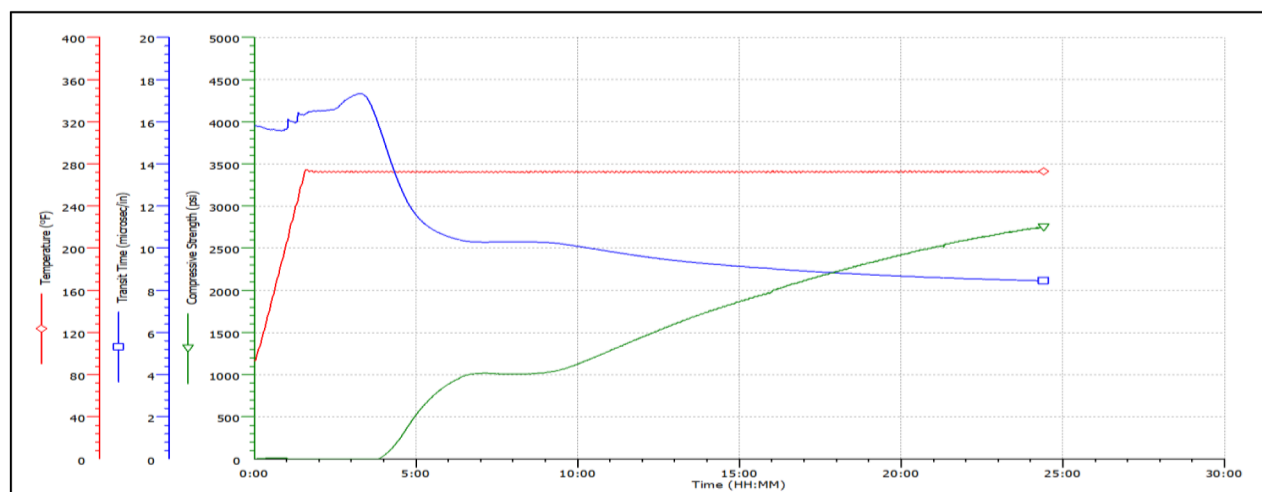


Figure 6—Compressive Strength Chart

After 24 hours of WOC, cement gain compressive strength to 2,732 psi as seen from the chart.

Cement Height Calculation. Considering the marginal depth between the perforation intervals, it was required to dump cement on the MPBT plug carefully in minimum dump bailer runs to achieve required cement height on the plug. Table-3 shows the height calculation achieved in each dump bailer run to achieve the desire cement height.

It has been observed from Table 7 that minimum $06 \times$ runs of the dump bailer required to achieve 10 ft height of the cement column on the MPBT plug. The MPBT plug and cement on it would give full isolation without impacting the upper zone perforations.

Table 7—Calculation of cement height & runs of dump bailer

Cement Height Calculation		
Casing ID	8.535	in
Casing Capacity	2.97	gal/ft
2-1/8" Bailer Capacity	0.13	gal/ft
Bailer Length	36	ft
Bailer Volume	4.82	gal
Casing Height	1.62	ft
Height Required	10	ft
Min. Number of Runs	6	Runs

Cement Dump Bailer Function & Capacity. Cement dump bailer is a tool that has a piston and holes at the bottom which operated electrically on wireline to dump cement. Dump bailer has the capacity of carrying and dumping ~ 5.0 gallons in $01 \times$ run. The schematic of the tool is shown below in Figure 7.

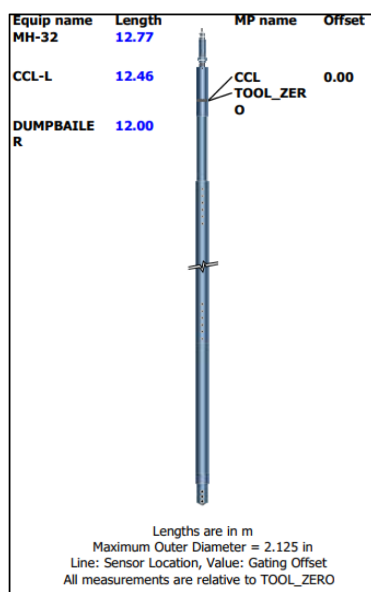


Figure 7—Schematic of Dump Bailer Tool

Cement was dumped in $6 \times$ dump bailer runs to achieve cement column of 10 ft, directly above the inflated plug to sustain differential pressure of 3500 psi, creating a robust zonal barrier that permanently isolated the water inflow. For quality control, a cement sample is collected from each bailer run and placed in a water bath to observe the curing behavior and strength development.

After wait on cement (WOC) to allow full hydration and setting, the cement column was tagged using a 2.70" OD gauge cutter to confirm the top of cement. This provided assurance that sufficient cement height is achieved, and the system integrity is intact.

This comprehensive procedure not only enhances plug anchoring confidence but also reinforces pressure isolation above the plug, making the well ready for shut-off verification or further intervention as required.

The same methodology was adopted for Well-Y. However, for Well-Z, due to large scale deposition and casing ovality, plug was set across the perforation at 2696.7 m followed by dumping of cement.

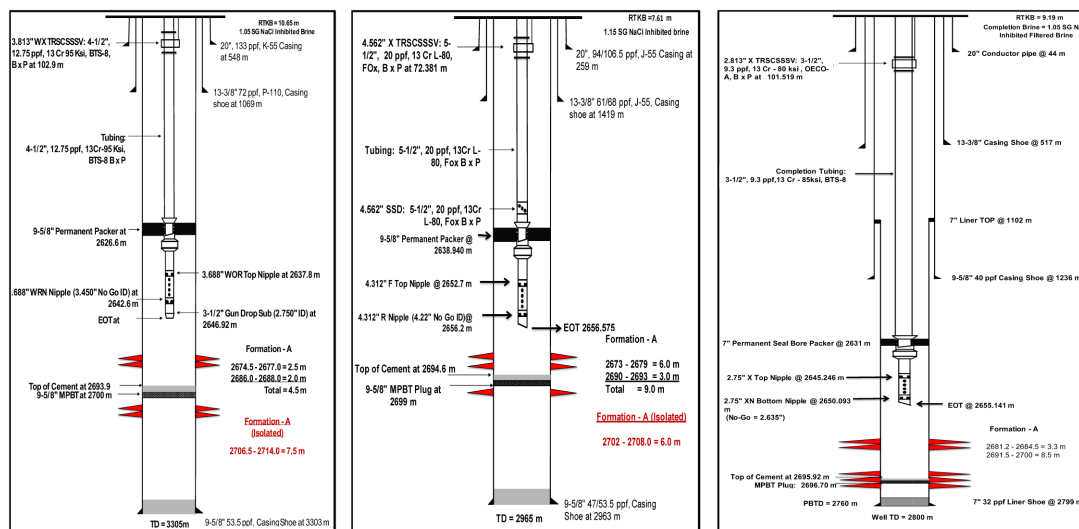


Figure 8—Wellbore Completion Sketch of Well X, Y & Z.

Results

Post-WSO shows a significant improvement in wellbore hydraulics with an increase in gas rates and well-head flowing pressures and reduction in the produced water fraction, improving gas velocity and drawdown efficiency. Significant achievement in the incremental gas production rates and the shut-off of high producing water zone was observed in Well X & Well Y. Both wells, were taken into the system and the performance of both well was monitored critically for the initial few days. It was observed that in Well X, 99% water cut was reduced along with 40% increment in WHFP. Similarly, in Well Y, 98% water cut was reduced along with 15% increment in WHFP was observed at similar rates.

The reduction in water fraction and the increment in gas rate significantly improved the life of the field and also assisted in water handling issues. Due to significant improvement in wellbore hydraulics, same approach is being considered across the field on different reservoirs to cater for the water handling issues, improve the integrity of surface facilities and reduce the requirement of water disposal wells. Figure 9 is showing a comparative analysis of water rate, gas rate & WHFP for pre & post water shut-off job performed on wireline through the rigless well intervention.

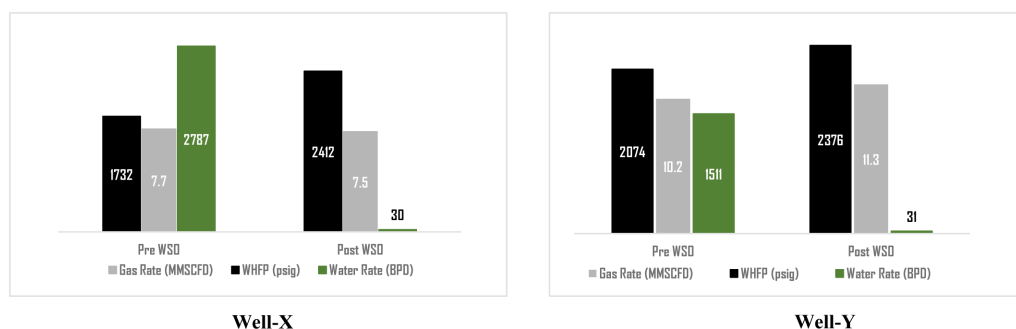


Figure 9—Comparison of Pre and Post Flowing Parameters.

Lessons Learned: Navigating Challenges for Continuous Improvement

In every successful operation, there are embedded lessons from past setbacks, and the rigless WSO campaign was no exception. At Well-X, while cement dumping above the plug using a dump bailer, after POOH of the second cement dump bailer run, pre-mature setting of cement was found inside dump bailer, a setback that threatened the integrity of the isolation. A critical analysis revealed issues with the original slurry formulation. In response, the cement recipe was revised by incorporating a retarder and adopting high-RPM mixing to ensure homogeneity and delay setting. This modified approach proved effective and was successfully replicated in Well-Y.

At Well-Z, another challenge emerged; severe casing ovality and heavy scale that limited the available space between perforation intervals for the plug placement. After a thorough production logging interpretation and gas-water contact (GWC) evaluation, the decision was made to set the inflatable plug across the bottom perforation set at 2696.7 m. Cement was then dumped above the plug based on volume calculations. However, post WOC, tagging revealed a total cement loss into the formation, inadvertently damaging the reservoir. This failure highlighted a critical operational insight: cement should not be dumped above plugs set across open perforations due to direct formation communication.

These experiences underscore a fundamental principle: each job, whether deemed a success or setback, adds to a reservoir of knowledge. Lessons learned from Well-X and Well-Z were pivotal in refining operational procedures, risk management, and treatment design, ultimately enhancing the overall reliability and effectiveness of rigless WSO applications in similar kinds of wells.

Conclusion

The integrated rigless WSO using inflatable thru-tubing plugs offers a very cost-effective solution in a sandstone reservoir in a mature field. By eliminating the need for rig workover and well killing, it preserves formation deliverability, minimizes cost, reduces deferment, and increase wellbore hydraulic. Proper screening, planning, and post-WSO monitoring are essential to ensure long-term success. While challenges persist in inflation control and deployment depth, continuous material advancements and digital monitoring are making this approach increasingly robust and scalable for mature gas assets.

By WSO, the risk of scale formation in the wellbore has been significantly reduced. This, in turn, mitigated the need for costly and time-consuming scale removal intervention jobs, leading to further operational savings. Elimination of handling of water at the facility substantially enhanced the operational life and integrity of the surface facilities. This also reduced the need for frequent maintenance and repair.

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NOMENCLATURE

- Water Shut-Offs (WSO)
- Pressure Control Equipment (PCE)
- Million Standard Cubic Foot per Day (MMSCFD)
- Casing Collar Locator (CCL)
- Cement Bond Log (CBL)
- Variable Density Log (VDL)
- Production Logging Tool (PLT)
- Coiled Tubing (CT)
- Wait on Cement (WOC)

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